Answers and talk notes

Represent and Describe Data in a Box Plot

What your student is learning

Tonight we put two box plots side by side and compare two groups — first by their centers, then by how spread out they are.

Answers

Ask your student to explain before you look.

1. **A.** The median line. The median line is the typical value, so compare the median lines.
2. **B.** Group B. 52 is greater than 40, so Group B is typically higher.
3. **C.** Class Y, because its box is narrower. Class Y has a narrower box (52 – 38 = 14 versus 60 – 30 = 30), so Class Y, because its box is narrower.
4. **D.** 16. The width is Q3 – Q1 = 41 – 25 = 16.

Talk about it

Which week has the higher median, and which has the narrower box?

Home mission

Think of two bus routes you know. Which is typically faster? Which is more on time every day? Can one route win one question and lose the other?

When they are stuck, ask

- Which median line sits higher on the scale?
- Which box is wider? What does the width tell us?
- Can you compare center and spread in the same sentence?

Try not to say

- The bigger box means the bigger numbers.
- Just look at which plot is longer overall.
- If the medians are the same, the groups are the same.